

Agentic Data Analyst

Eesha Kulkarni

Agenda



PROBLEM



SOLUTION



LOOPHOLES



FUTURE WORK

BEFORE:
Manual, Tedious



**CONFUSION &
OVERLOAD**



Error
Prone



Repetitive
Tasks



Poor
Outcomes



Automated
Insights

AFTER:
AI-Driven, Efficient



Interactive
Dashboard



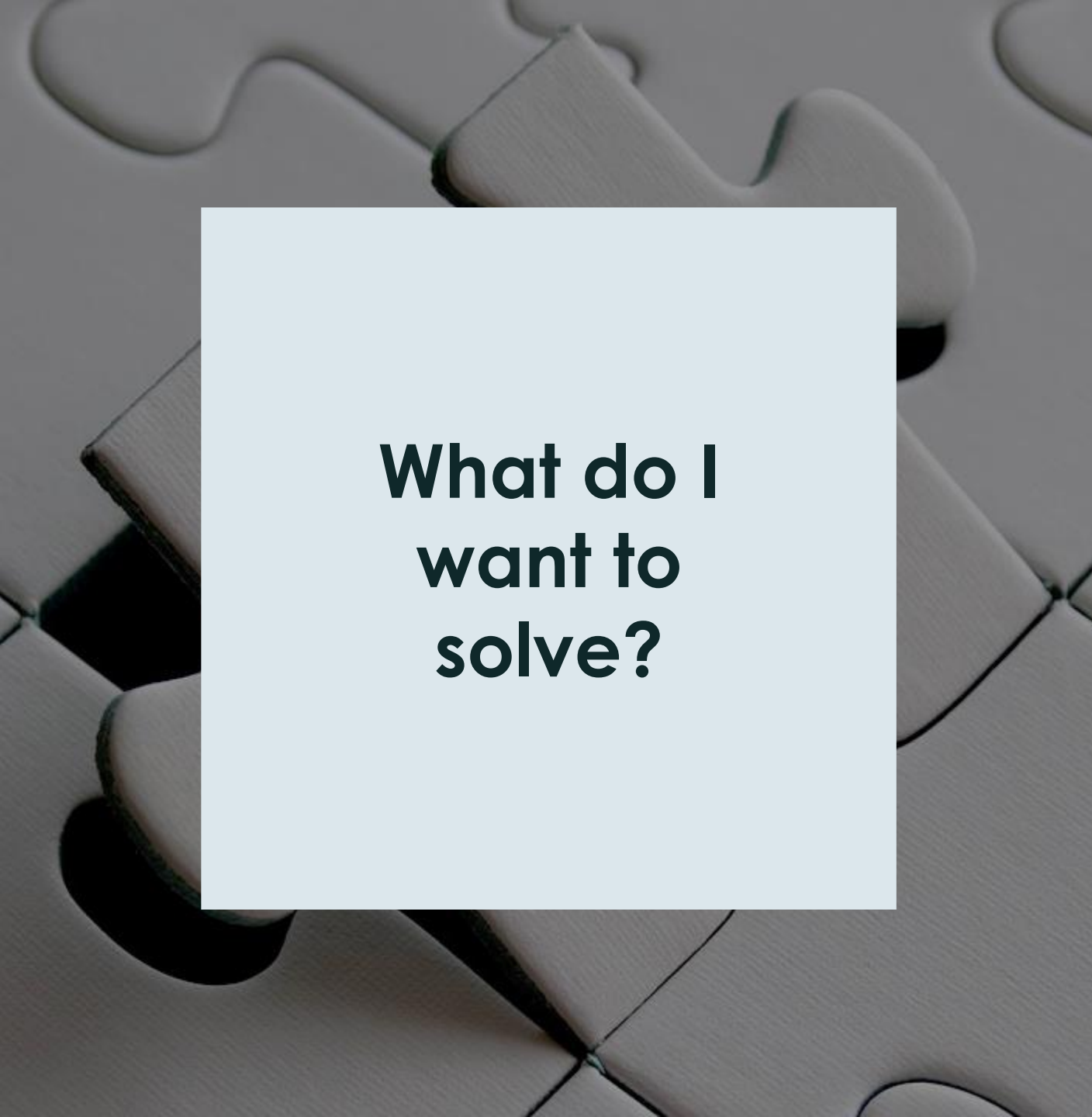
Insights



Automated
Reports

Motivation

- Met Pratiti at an event - got to learn what Sundial does
- Conversation with a mentor at my previous internship
- Experience creating dashboards in Tableau
- How about creating my own version of an AI analyst?



What do I want to solve?

- How can I use AI to assist with the tedious dashboarding process?
- Can I modify the AI given dashboard easily without errors?
- Is human-in-the-loop my design or a fallback?

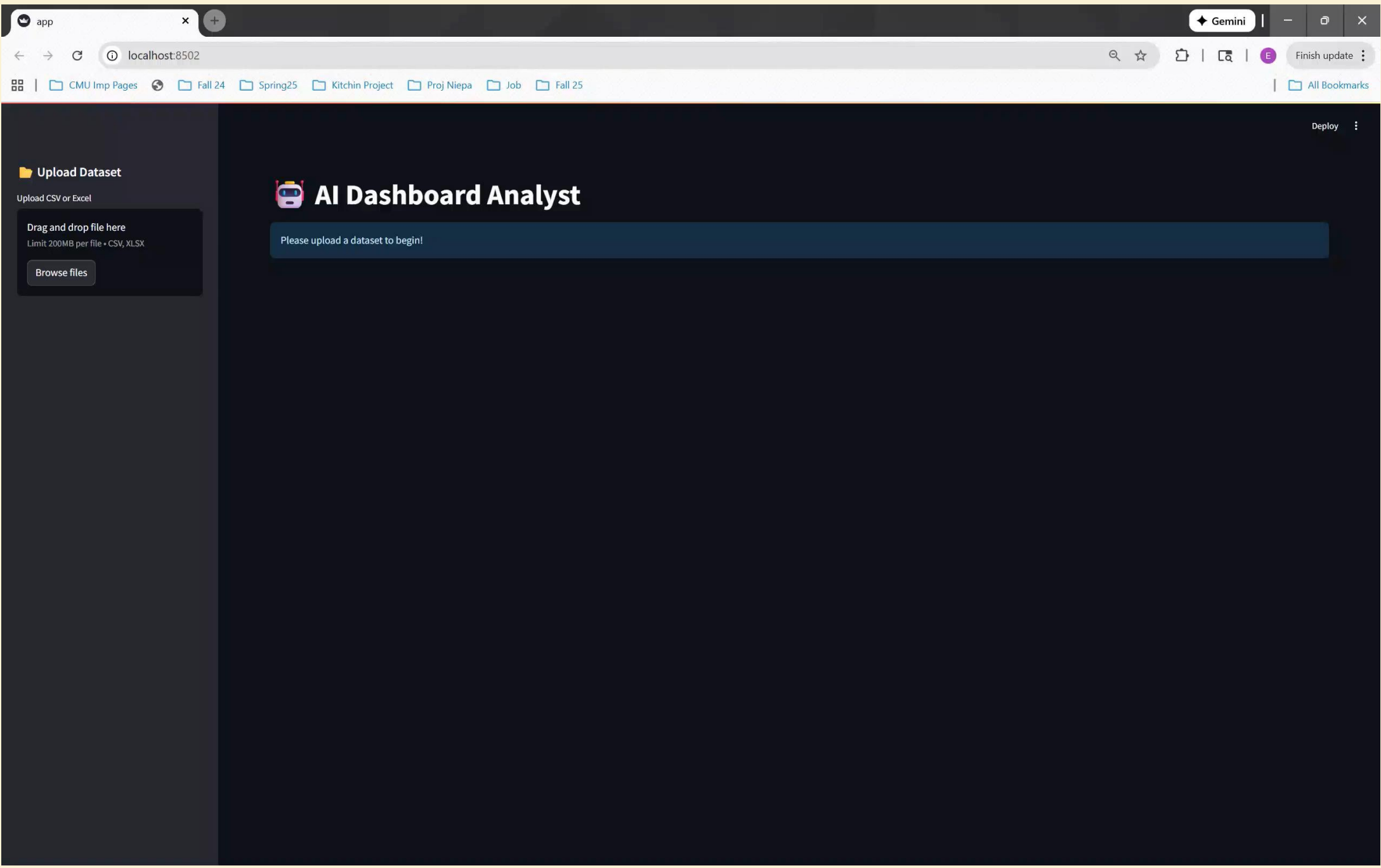
AI Dashboard Analyst

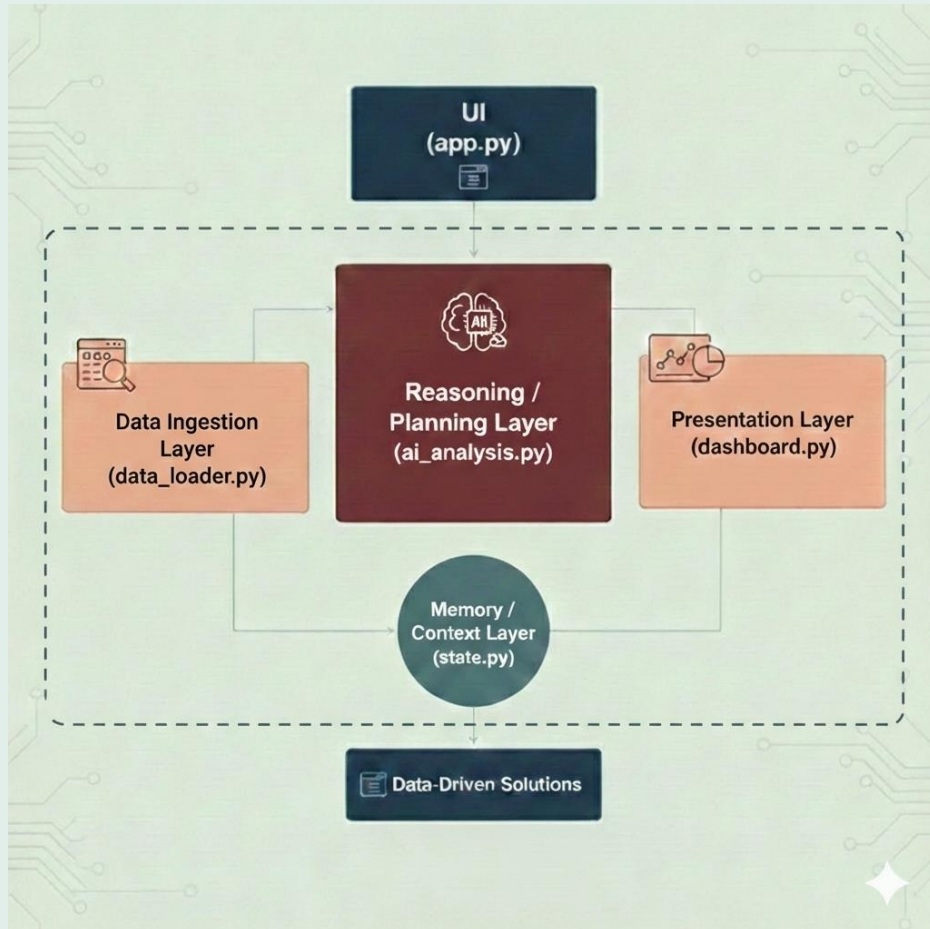
Date: 10/10/2023 Insights

Chat History

THE SOLUTION

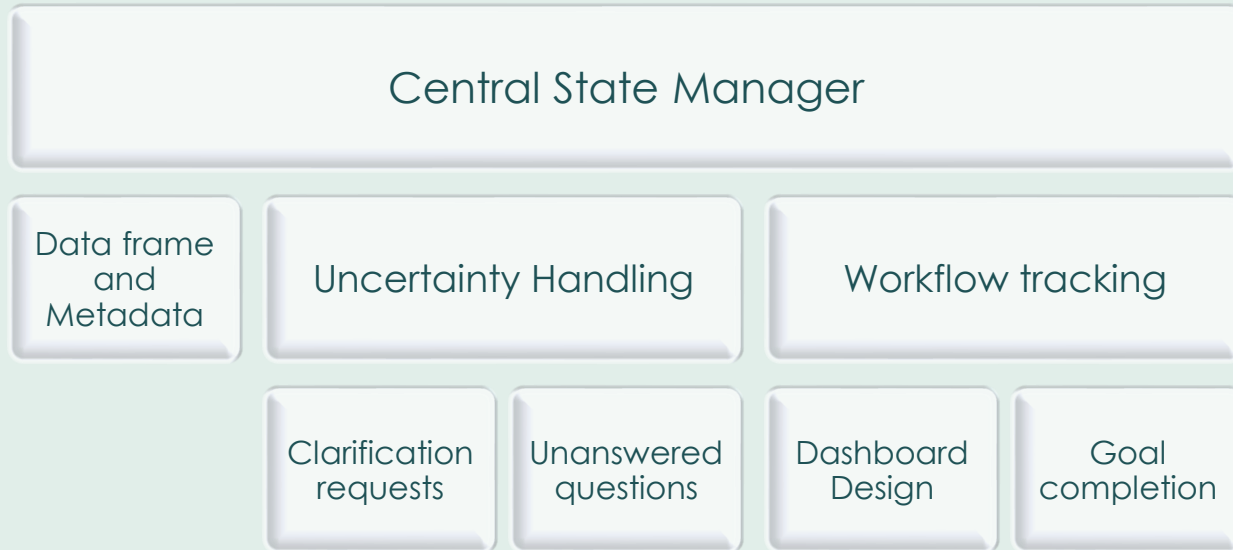




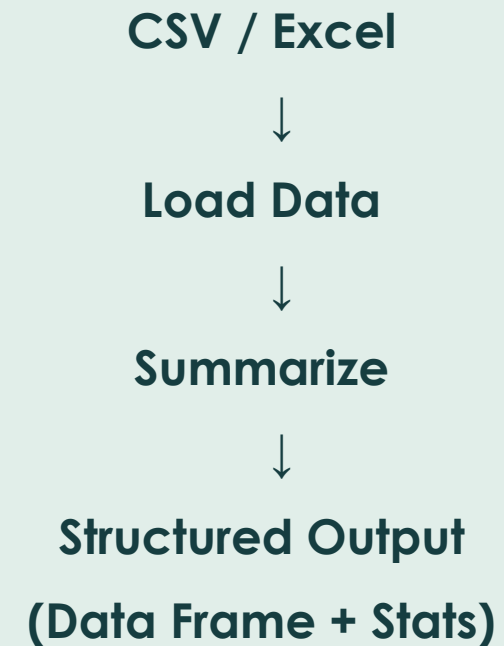


Structuring my approach

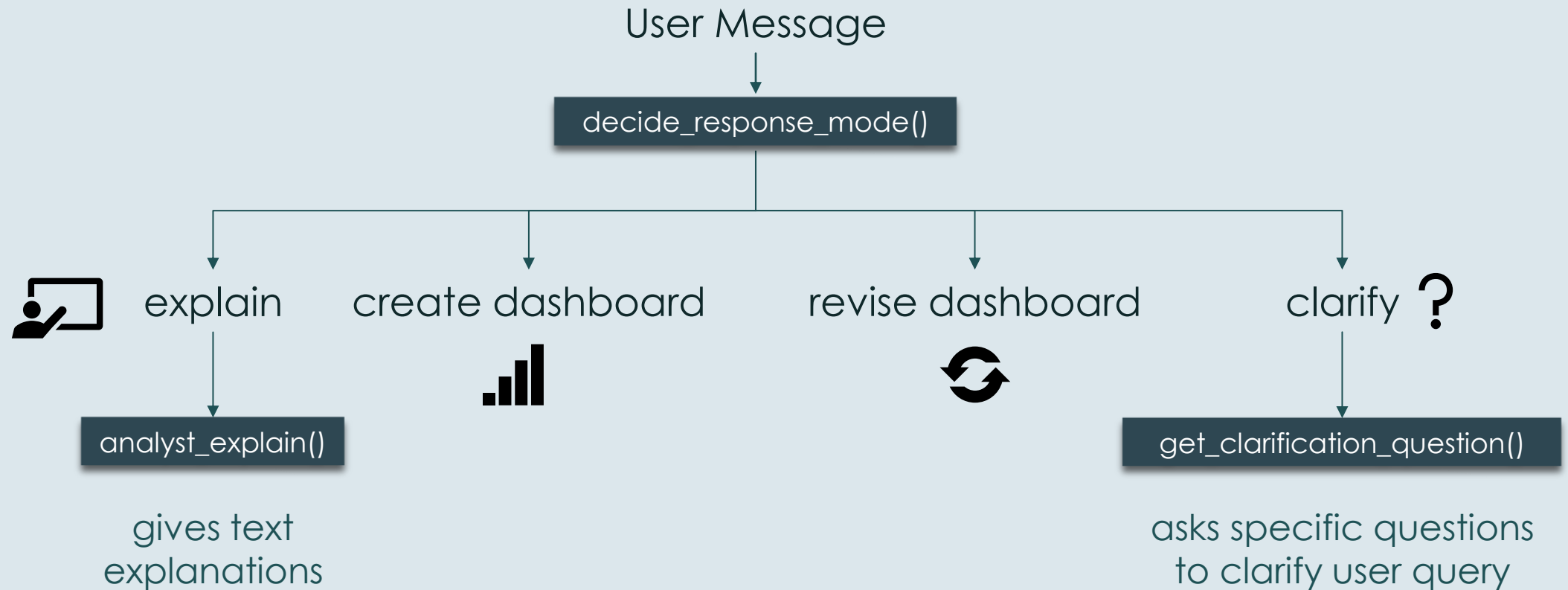
state.py



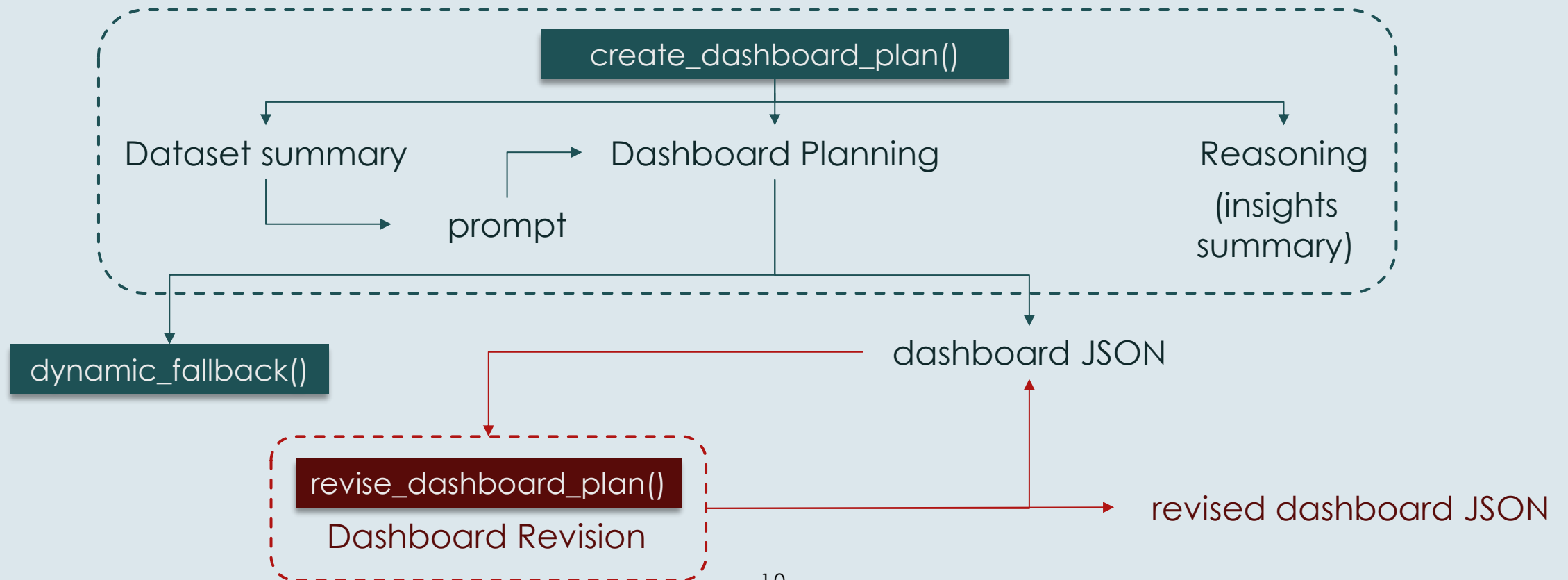
data_loader.py



ai_analysis.py – intent & control

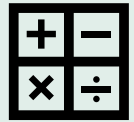


ai_analysis.py – planning & guardrails



dashboard.py

render_kpis()



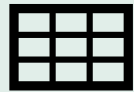
math



format

KPI

render_filters()



column
based



user
slicing

Filters

render_dashboard()



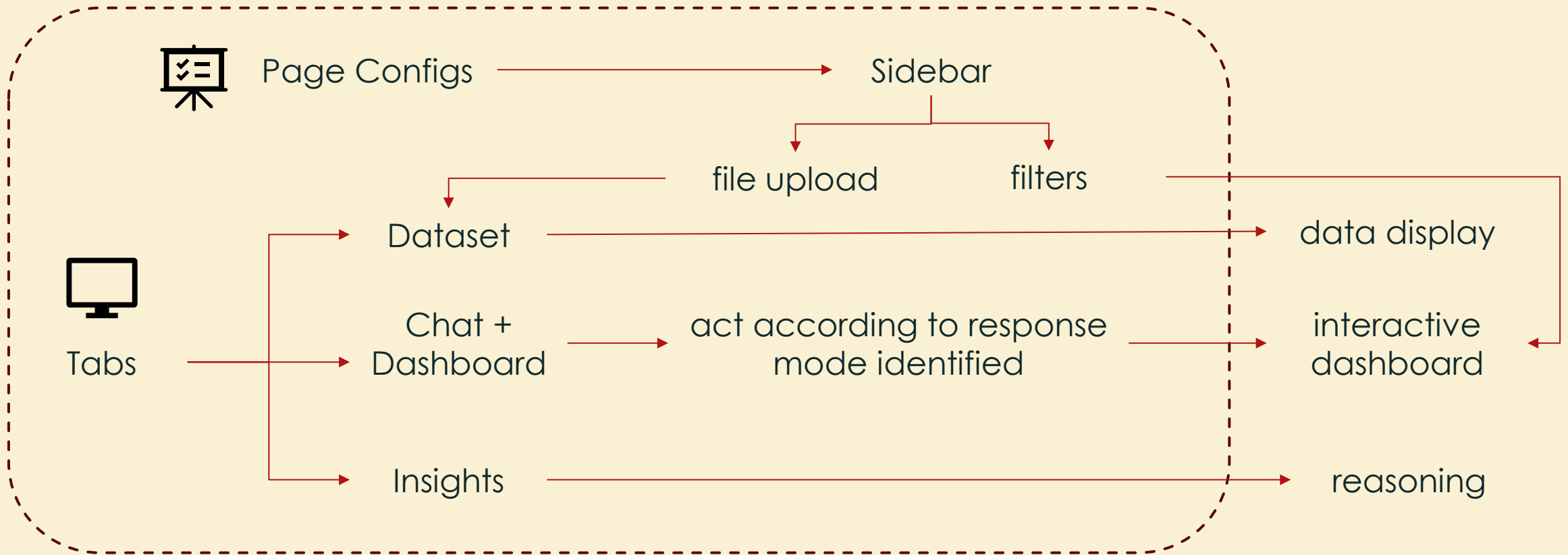
layout



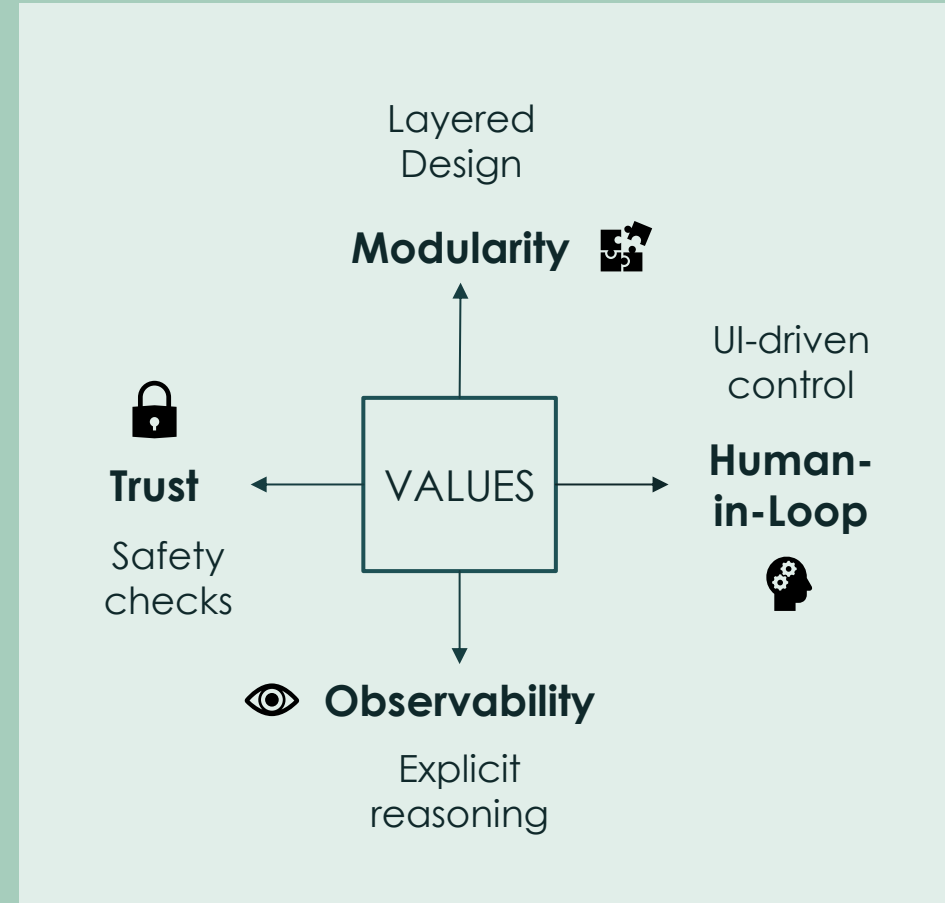
plan-driven

Dashboard

app.py



How does it align with Sundial?



What I'd fix with more resources and time...



Single-User Focus

Why it's a Problem

State isolated to one user session.
No collaboration or history.



No Real-Time Input

Why it's a Problem

Cannot process data streams.
Insights are not current.



Basic Charts Only

Why it's a Problem

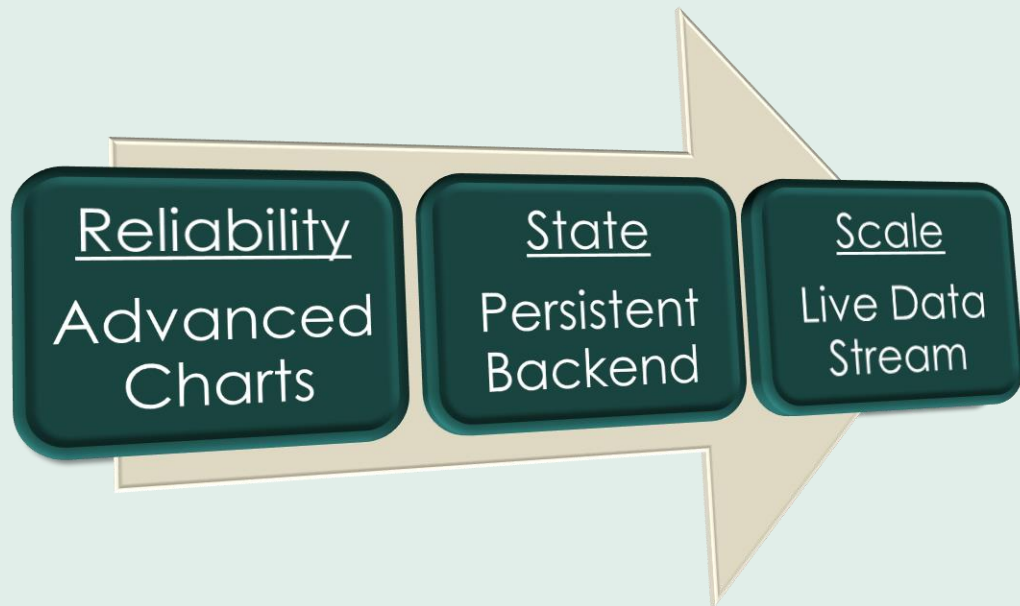
Fixed set of simple visualizations.
Limited insights from data.



Context Window Limits

Why it's a Problem

Input size is token-constrained.
Limited data / convo history.



**Plan
ahead...**

THANK YOU

QUESTIONS?